

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : D9

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1. Problem Title : Library Management System

2. Problem Description :

At the start of a semester, students rush to borrow textbooks. Instead of writing entries in a register, the librarian opens the system and searches for the requested book. When issuing, the librarian enters the student ID and book ID. The system checks availability. If available, it creates a borrow record and marks the book unavailable. Students later log in to see which books they currently hold and their due dates. During returns, the librarian processes the transaction, and the system updates CSV records. Over time, the library maintains accurate borrowing histories. The system prevents issuing already-borrowed books and helps identify overdue returns, improving discipline and efficiency

3. Identified Entities (Classes)

Class Name	Description
Book	Information regarding book and availability.
Student	Information regarding student and methods to view.
Library	The Library class manages the operations such as adding books, issuing and returning books, and maintaining borrowing records.
Main	The Main class handles user interaction, displays menus, and controls the overall flow.

FileHandler	The purpose of this class is to define methods to write, read and append to the files required by the application.
BorrowRecord	The class contains the details of the books borrowed, by which students, the date the book was issued and the due data.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Book	-bookId : String -title : String -author : String -isAvailable : boolean	+getBookId() : String +getTitle() : String +getAuthor() : String +isAvailable() : Boolean +setAvailable(Boolean available) : void +toString() : String
Student	-studentId : String -name : String	+getStudentId() : String +getName() : String +toString() : String
Library	- <u>BOOKS_FILE</u> : String - <u>BORROW_FILE</u> : String	+viewBooks() : void +addBook(Book book) : void +issueBook(String studentId, String bookId) : void +returnBook(String bookId) : void +viewIssuedBookByStudent(String studentId) : void
FileHandler	Nil	+ <u>readCSV(String filename) : List<String[]></u> + <u>writeCSV(String filename, List<String> lines) : void</u> +appendCSV(String file
BorrowRecord	-studentId : String -bookId : String -issueDate : String -dueDate : String	+toString() : String

